Experiment-5

Aim: Exercise on retrieving records from the table.

EMPLOYEES (Employee_Id, First_Name, Last_Name, Email, Phone_Number, Hire_Date, Job_Id, Salary, Commission_Act, Manager_Id, Department_Id).

- **Create Database and Table**
  - **Code:**

```
▷Run
CREATE DATABASE ASSIGNMENT_5;

▷Run
USE ASSIGNMENT_5;

▷Run | Select
CREATE TABLE EMPLOYEES (
    Employee_Id INT PRIMARY KEY,
    First_Name VARCHAR(50),
    Last_Name VARCHAR(50),
    Email VARCHAR(100),
    Phone_Number VARCHAR(15),
    Hire_Date DATE,
    Job_Id VARCHAR(20),
    Salary DECIMAL(10,2),
    Commission_Act DECIMAL(5,2),
    Manager_Id INT,
    Department_Id INT
);
```

  - **Explanation:**
  - This creates a new database named *ASSIGNMENT_5*.
  - This command tells SQL to use the created database.
  - This creates a table with employee details like ID, name, salary, department, etc.

- **Insert Sample Data**
- **Code:**

```sql
INSERT INTO EMPLOYEES VALUES
(111, 'Varun', 'King', 'SKING', '1234567890', '2003-06-17', 'AD_PRES', 24000, NULL, NULL, 90),
(121, 'Neena', 'Kochhar', 'NKOCHHAR', '1234567891', '2005-09-21', 'AD_VP', 17000, NULL, 100, 90),
(122, 'Lex', 'De Haan', 'LDEHAAN', '1234567892', '2001-01-13', 'AD_VP', 17000, NULL, 100, 90),
(123, 'Alexander', 'Hunold', 'AHUNOLD', '1234567893', '2006-01-03', 'IT_PROG', 9000, NULL, 102, 60),
(124, 'Bruce', 'Austin', 'BAUSTIN', '1234567894', '2007-05-21', 'IT_PROG', 6000, NULL, 103, 60),
(125, 'David', 'Austin', 'DAUSTIN', '1234567895', '2005-06-25', 'IT_PROG', 4800, NULL, 103, 60),
(126, 'Valli', 'Pataballa', 'VPATABAL', '1234567896', '2006-02-05', 'IT_PROG', 4800, NULL, 103, 60),
(127, 'Diana', 'Lorentz', 'DLORENTZ', '1234567897', '2007-02-07', 'IT_PROG', 4200, NULL, 103, 60);
```

- **Explanation:**
  - Inserts sample employee data for testing queries.

(a) Find out the employee id, names, salaries of all the employees

- **Code:**

```sql
SELECT Employee_Id, First_Name, Last_Name, Salary
FROM EMPLOYEES;  6ms
```

- **Output:**

Employee_I int	First_Name varchar(50)	Last_Name varchar(50)	Salary decimal(10,2
100	Steven	King	24000.00
101	Neena	Kochhar	17000.00
102	Lex	De Haan	17000.00
103	Alexander	Hunold	9000.00
104	Bruce	Austin	6000.00
105	David	Austin	4800.00
106	Valli	Pataballa	4800.00

- **Explanation:**
  - SELECT chooses specific columns
  - Displays employee ID, first name, last name, and salary

(b) List out the employees who works under manager 100

- **Code:**

```
SELECT *
FROM EMPLOYEES
WHERE Manager_Id = 100;   5ms
```

- **Output:**

101	Neena	Kochhar	NKOCHHAR	1234567891	2005-09-21	AD_VP	170
102	Lex	De Haan	LDEHAAN	1234567892	2001-01-13	AD_VP	170
121	Neena	Kochhar	NKOCHHAR	1234567891	2005-09-21	AD_VP	170
122	Lex	De Haan	LDEHAAN	1234567892	2001-01-13	AD_VP	170

➢ **Explanation:**

- WHERE filters records
- Shows employees whose manager ID is 100

(c) Find the names of the employees who have a salary greater than or equal to 4800

- **Code:**

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Salary >= 4800;   7ms
```

- **Output:**

First_Name varchar(50)	Last_Name varchar(50)
Steven	King
Neena	Kochhar
Lex	De Haan
Alexander	Hunold
Bruce	Austin
David	Austin
Valli	Pataballa

➢ **Explanation:**

- >= means greater than or equal
- Filters employees earning 4800 or more

(d) List out the employees whose last name is ‘AUSTIN’

- **Code:**

```
▷Run | +Tab | JSON
SELECT *
FROM EMPLOYEES
WHERE Last_Name = 'Austin';   4ms
```

- **Output:**

Employee_I int	First_Name varchar(50)	Last_Name varchar(50)	Email varchar(10C	Phone_Number varchar(15)	Hire_Date date	Job_Id varchar(2C
104	Bruce	Austin	BAUSTIN	1234567894	2007-05-21	IT_PROG
105	David	Austin	DAUSTIN	1234567895	2005-06-25	IT_PROG
124	Bruce	Austin	BAUSTIN	1234567894	2007-05-21	IT_PROG
125	David	Austin	DAUSTIN	1234567895	2005-06-25	IT_PROG

➢ Explanation:

- Matches exact text
- Displays employees with last name AUSTIN

(e) Find the names of the employees who works in departments 60,70 and 80

- **Code:**

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Department_Id IN (60, 70, 80);
```

- **Output:**

First_Name varchar(50)	Last_Name varchar(50)
Alexander	Hunold
Bruce	Austin
David	Austin
Valli	Pataballa
Diana	Lorentz
Alexander	Hunold
Bruce	Austin

➢ **Explanation:**

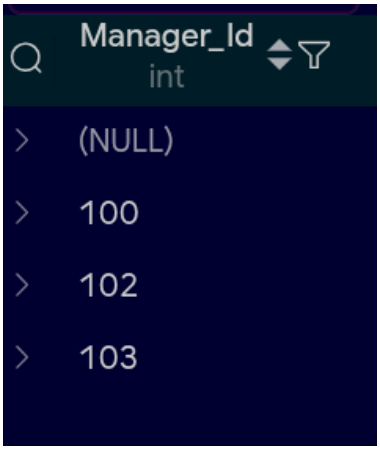
- IN is used to match multiple values
- Shows employees belonging to given departments

(f) Display the unique Manager_Id.

- **Code:**

```
SELECT DISTINCT Manager_Id
FROM EMPLOYEES;   5ms
```

- **Output:**



Manager_Id int
(NULL)
100
102
103

➢ **Explanation:**

- DISTINCT removes duplicate values
- Shows only unique manager IDs